

**BACHELOR OF COMPUTER
APPLICATIONS (BCA) (REVISED)**

Term-End Examination

December, 2025

BCS-040 : STATISTICAL TECHNIQUES

Time : 2 Hours

Maximum Marks : 50

Note : (i) Attempt both Sections i.e. Section A and Section B.

(ii) Attempt any **four** questions from Section A.

(iii) Attempt any **three** questions from Section B.

(iv) Use of non-scientific calculator is allowed.

Section—A

1. Calculate the correlation coefficient and slope of the regression line Y on X, by using the data given below : 5

$$\Sigma x = 15$$

$$\Sigma y = -6$$

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P. T. O.

[2]

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$$\Sigma xy = 50$$

$$\Sigma x^2 = 61$$

$$\Sigma y^2 = 90$$

Above data is calculated from 20 pairs of observations for variables X and Y.

2. Suppose A and B are two independent events, associated with a random experiment. If the probability of occurrence of either A or B is 0.6; while probability that only A occurs is 0.4, then determine the probability of occurrence of event B. 5
3. Suppose 2% of the items made in a factory are defective. Find the probability that there are : 5
- (i) 3 defectives in a sample of 100
- (ii) no defectives in a sample of 50
4. The mean weekly sales of chocolate packets in different departmental stores was 146.3 packets per store. After an advertisement campaign the mean weekly sales of 22 stores for a typical week increased to 153.7 and showed a standard deviation of 17.2. Was the

advertisement campaign successful at 5% level of significance ? (Given $t_{21(0.05)} = 2.08$). 5

5. Differentiate between the following : 5
- Parametric and Non-Parametric tests
 - Binomial distribution and Poisson's distribution
6. An incomplete frequency distribution is given below :

Class Interval	Frequency
10—20	12
20—30	30
30—40	?
40—50	65
50—60	?
60—70	25
70—80	18

Given that median value of 200 observations is 46. Determine the missing frequencies using the median formula. 5

P. T. O.

Section—B

7. Using the regression line :

$$\hat{y} = 90 + 50x$$

fill up the values in the table below : 10

Sample No. (i)	12	21	15	1	24
x_i	0.96	1.28	1.65	1.84	2.35
y_i	138	160	178	190	210
$\hat{y}_i$	138	—	—	—	—
$e_i = y_i - \hat{y}_i$	0	—	—	—	—

After filling the table, compute the parameters R and R^2 . Interpret the correlation between X and Y.

8. A chemical firm wants to determine how four catalysts (i.e. 1, 2, 3, 4) differ in their yields ? The firm conducted the experiment in three of its plants (i.e. A, B and C). In each plant, the

yield is measured with each catalyst. The yield (in quintals) is tabulated below : 10

Plant	Catalyst			
	1	2	3	4
A	2	1	2	4
B	3	2	1	3
C	1	3	3	1

Perform an ANOVA and comment whether the yield due to a particular catalyst is significant or not at 5% level of significance. Given $F_{3,6} = 4.76$.

9. What do you understand by the term 'Forecasting'? How does forecasting differ from prediction? Give suitable examples in support of your answer. Briefly discuss the forecasting models. 10
10. Telephone directories have telephone numbers which are the combinations of ten digits 0 to 9. The observer notes the frequency of occurrence of these digits and wants to test whether the

P. T. O.

digits occur with same frequency or not ($\alpha = 0.05$). The data is tabulated below :

Digits	Frequency
0	99
1	100
2	82
3	65
4	50
5	77
6	88
7	57
8	82
9	30

(Given that $\chi^2_{9(0.05)} = 16.918$). 10

BACHELOR OF COMPUTER APPLICATIONS

(BCA) (Revised)

Term-End Examination, 2019

BCS-040 : STATISTICAL TECHNIQUES

Time : Two Hours]

[Maximum Marks : 50

Note : Attempt both sections, i.e., Section A and Section B.

Attempt **any four** questions from **Section-A**. Attempt **any three** questions from **Sections-B**. Non-scientific calculator is allowed.

SECTION - A

1. The following table represent daily wages (in rupees) of the workers in a certain commercial organisation : [5]

Daily Wages	200-300	300-400	400-500	500-600
Number of Workers	05	10	03	02

Calculate mean and standard deviation of the above data.

2. A person is known to hit a target in 4 out of 5 shots whereas another person is known to hit in 2 out of 3 shots. Find the probability that the target being hit, when they both try. [5]
3. An oil exploration firm plans to drill four holes. It is assumed that the probability of oil yield from each hole is $1/5$. Since the holes are in quite different locations, the outcome of drilling one holes is statistically independent of that of drilling any other holes. What is the probability that two or more holes produce oil. [5]
4. A population of five households having monthly income (in thousand rupees) as following :

Household	1	2	3	4	5
Income	14	10	12	20	18

Write all possible samples without replacement of size 2 and show that sample mean gives an unbiased estimate of population mean.

5. Explain **any two** of the following : [5]
 - (a) z-test for mean
 - (b) Paired t-test

- (c) Simple random sampling

SECTION - B

6. An investigator is interested to know the level of knowledge about the history of India, among three different schools in a city. A test is given to four students of 8th class of each school. Their scores out of 10 are given below :

School I	School II	School III
8	6	6
6	4	5
7	6	5
5	5	6

Test the equality of the average scores of the three schools at 5% level of significance. [10]

(Given $F_{(2,9)} 5\% = 4.26$)

7. A Manager of a car company wants to estimate the relationship between age of cars and annual maintenance cost. The following data from six cars of same model are obtained as :

Age of Car (in years)	Annual Maintenance Cost (In hundred rupees)
1	10
2	15
3	18
4	20
5	25
6	35

- (a) Construct a scatter diagram for the data given above.
- (b) Fit a best linear regression line, by considering annual maintenance cost as the dependent variable and the age of the car as the independent variable.
- (c) Use this regression line to predict the annual maintenance cost for the car of age 8 years.

[2+7+1]

8. The following contingency table presents the analysis of 300 persons according to hair colour and eye colour :

Hair Colour	Eye Colour			Total
	Blue	Grey	Brown	
Fair	30	10	40	80
Brown	40	20	40	100
Black	50	30	40	120
Total	120	60	120	300

Test the hypothesis that there is an association between hair colour and eye colour at 1% level of significance

(Given $\chi^2_{(4), 1\%} = 13.28$, $\chi^2_{(6), 1\%} = 16.81$) [10]

9. (a) Define stratified random sampling.
- (b) Suppose the population of three towns is $N_1 = 50000$, $N_2 = 30000$, and $N_3 = 40000$, respectively. A stratified random sample is to be drawn with a total sample size of $n = 500$. Determine the sample size for each town individually using the method of :
- proportional allocation
 - Optimal allocation.

It is known from a previous survey that $S_1 = 30$, $S_2 = 15$ and $S_3 = 20$

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**BACHELOR OF COMPUTER APPLICATION
(BCA) (Revised)**

Term-End Examination

BCS-040 : STATISTICAL TECHNIQUES

Time : 2 Hours]

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Attempt any four questions from Section-A. Attempt
any three questions from Section-B. Use of non-
scientific calculator is allowed.

Section-A

1. Given the following sample of 20 numbers: 5
12, 41, 48, 58, 14, 43, 50, 59, 15, 45, 52, 72, 18,
45, 54, 78, 41, 47, 56, 79
- (i) Compute mean, variance and standard deviation.
- (ii) If the largest value in the sample of 20 number given above, is changed to 500, then to what extent the mean and variance will change? Justify your answer.



2. A dice is rolled 1200 times with the following results: 5

No. that comes up	1	2	3	4	5	6
Frequency	195	289	202	242	163	109

Test the hypothesis, if the dice is unbiased at 5% level of significance (Given that $\chi^2_{0.05}(5) = 11.07$)

3. Calls at a telephone switchboard occur at an average rate of 6 calls per 10 minutes. Suppose the operator leaves for a 5 minutes coffee break. What is the probability that exactly two calls occur while the operator is away? 5
4. Fit a linear trend $y = a + b * (\text{Demand})$, to the data collected from an umbrella manufacturing unit: 5

Month	1	2	3	4	5	6
Demand	46	56	54	43	57	56

5. Construct ANOVA table for one-way classification. 5
6. Briefly discuss, any two of the following: 5
- (i) Goodness of fit test
 - (ii) Binomial distribution
 - (iii) t-test for mean

Section-B

7. In a partially destroyed laboratory, legible record for correlation analysis of data is preserved as follows:

(a) Variance of $x = 9$

(b) Regression equations:

(i) $8x - 10y + 66 = 0$

(ii) $40x - 18y - 214 = 0$

Analyse the preserved records and determine:

- (i) The mean of x and y .
(ii) The coefficient of correlation between x and y .
(iii) The standard deviation of y .

10

8. The following table shows the sample values of 3 independent normal random variables i.e. X_1 , X_2 and X_3 . Assuming that they have equal variance, test the hypothesis that they have the same mean,

by using ANOVA (Given $F_{(2, 9)}^{(0.05)} = 4.26$) 10

X_1 :	13	11	16	22
X_2 :	16	8	21	11
X_3 :	15	12	25	10

9. What do you understand by the term "Time Series"?
Discuss all the categories in which Time Series is classified. 10
10. Discuss the term "Systematic Sampling".
Differentiate between Linear and Circular systematic sampling. Give two advantages and limitations of systematic sampling. 10

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No. of Printed Pages : 6

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**BACHELOR OF COMPUTER
APPLICATIONS (BCA) (REVISED)**

Term-End Examination

December, 2020

BCS-040 : STATISTICAL TECHNIQUES

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Section—A

1. Calculate the correlation coefficient and slope of the regression line Y on X, by using the data given below : 5

$$\Sigma x = 15$$

$$\Sigma y = -6$$

$$\Sigma xy = 50$$

$$\Sigma x^2 = 61$$

$$\Sigma y^2 = 90$$

Above data is calculated from 20 pairs of observations for variables X and Y.

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advertisement campaign successful at 5% level of significance ? (Given $t_{21(0.05)} = 2.08$). 5

5. Differentiate between the following : 5

- (i) Parametric and Non-Parametric tests
- (ii) Binomial distribution and Poisson's distribution

6. An incomplete frequency distribution is given below :

Class Interval	Frequency
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Given that median value of 200 observations is 46. Determine the missing frequencies using the median formula. 5

Section—B

7. Using the regression line :

$$\hat{y} = 90 + 50x$$

fill up the values in the table below : 10

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The firm conducted the experiment in three of its plants (i.e. A, B and C). In each plant, the

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Plant	Catalyst			
	1	2	3	4
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C	1	3	3	1

Perform an ANOVA and comment whether the yield due to a particular catalyst is significant or not at 5% level of significance. Given $F_{3,6} = 4.76$.

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Digits	Frequency
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1	100
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4	50
5	77
6	88
7	57
8	82
9	30

(Given that $\chi^2_{9(0.05)} = 16.918$).

10

**BACHELOR OF COMPUTER APPLICATIONS
(BCA) (Revised)**

Term-End Examination

June, 2021

BCS-040 : STATISTICAL TECHNIQUES

Time : 2 hours

Maximum Marks : 50

Note :

- (i) Attempt both sections, i.e., Section A and Section B.
- (ii) Attempt any **four** questions from Section A.
- (iii) Attempt any **three** questions from Section B.
- (iv) Use of non-scientific calculator is allowed.

SECTION A

1. An Automobile service centre performed the following number of car services each month :

15, 18, 25, 40, 25, 18, 25, 21, 30, 33, 25, 20, 10, 28,
36, 15, 26, 35, 20, 21, 32, 40, 32, 16, 12, 14, 22, 26,
37, 16.

- (i) Construct a continuous frequency distribution of the above data by taking suitable class width.
- (ii) Draw the histogram of constructed continuous frequency distribution in part (i).

5

2. The information regarding production of wheat (in thousand kg) in 25 districts is collected for a particular season. Select a possible systematic random sample of 7 units from the data given below :

23, 20, 30, 37, 76, 36, 13, 36, 16, 58, 53, 83, 10, 15, 13, 17, 12, 16, 17, 21, 20, 18, 61, 31, 71.

Also calculate the sample mean from the selected units.

5

3. A shopkeeper claims that the average life of a CFL Bulb is 1600 hours. To check this claim, a researcher takes a sample of 100 CFL bulbs of the same make randomly and finds mean lifetime of 1570 hours with standard deviation of 120 hours. Is the claim acceptable at 5% level of significance ? [Given that $Z = \pm 1.96$]

5

4. The incidence of occupational disease in an industry is such that the workers have a 20% chance of suffering from it. What is the probability that out of six workers

- (i) Four or more will contract the disease ?
- (ii) Three will contract the disease ?

5

5. Differentiate between the following with one example of each :

- (i) Assignable causes and Chance causes
- (ii) Producer's risk and Consumer's risk

5

6. Use moving average of length 3 for the data on sales given in following table :

<i>Month</i>	<i>Sales</i>
1	25
2	15
3	30
4	38
5	58
6	62
7	85
8	88
9	60
10	40
11	40
12	38

Calculate the moving average and plot it against the appropriate month. Also plot the data in the same graph.

5

SECTION B

7. A researcher wishes to test whether a person's cholesterol level will change if the diet is changed. She takes a sample of 15 persons and divides them into 3 diet groups randomly. The data of cholesterol levels (in mg/deciliter) for the three groups are given below :

<i>Cholesterol Levels</i>		
<i>Diet 1</i>	<i>Diet 2</i>	<i>Diet 3</i>
190	210	170
170	200	200
210	230	210
180	210	200
200	190	205

Test whether the average effect of different diets on cholesterol level differs significantly at 5% level of significance.

[Given that $F_{(2, 12)}(0.05) = 3.885$]

10

8. The following contingency table presents the analysis of 300 persons according to hair colour and eye colour :

<i>Hair Colour</i>	<i>Eye Colour</i>		
	<i>Blue</i>	<i>Grey</i>	<i>Brown</i>
Burgundy	30	10	40
Brown	40	20	40
Black	50	30	40

Test the hypothesis that there is an association between hair colour and eye colour at 5% level of significance. [Given that $\chi^2_{4, 0.05} = 9.48$]

10

9. Determine a regression equation $y = a + bx$ for the data given in the table below :

x	y
1	2
1	3
2	5
3	6
4	8
4	8
5	9
6	10
6	9
7	11

Use fitted regression line to predict y when $x = 8$.

Draw a scatter diagram to represent the data. 10

10. Write short notes on any **two** of the following : $5 \times 2 = 10$

- (i) Exponential Smoothing Method
 - (ii) Cluster Sampling
 - (iii) Correlation and Rank-correlation
-

No. of Printed Pages : 7

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**BACHELOR OF COMPUTER
APPLICATIONS (BCA) (REVISED)**

Term-End Examination

December, 2021

BCS-040 : STATISTICAL TECHNIQUES

Time : 2 Hours

Maximum Marks : 50

Note : (i) Attempt both Sections i.e. Section A and Section B.

(ii) Attempt any **four** questions from Section A.

(iii) Attempt any **three** questions from Section B.

(iv) Use of non-scientific calculator is allowed.

Section—A

1. A pharmacologist measured the amount of dopamine (no. of moles/gm) in the brains of

each of 10 rats. The observed data is given below :

6.8, 7.0, 5.3, 6.0, 5.9, 5.6, 6.8, 7.4, 6.2, 7.0

Calculate :

5

(i) The mean and standard deviation

(ii) Coefficient of variation

2. In a certain population of women, 5% have breast cancer, 15% are smokers and 3% are smokers as well as have breast cancer. A woman is selected and random from the population. What is the probability that she has breast cancer or she is a smoker ?

5

3. The pulse rate of 6 people was recorded before and after taking a new drug, which are given below :

	Pulse Rate					
Before Drug	68	71	84	93	67	74
After Drug	71	70	82	97	73	80

On the basis of the above information, can you conclude that there is a significant effect of the

[3]

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new drug to increase the pulse rate at 1% level of significance ? 5

[Given that $t_{5(0.01)} = 4.032$]

4. A company produces electric bulbs. To estimate the average life of the bulbs, the quality inspector selects 80 bulbs randomly. She finds that the average life of these bulbs is 50 hours. Find the population average life of the electric bulbs produced by the company using 95% confidence limits. 5

5. The average yield of wheat crop (in quintals/ha) pertaining to fields of 40 villages in a district is given below :

20, 21, 32, 41, 55, 22, 64, 42, 28, 35, 25, 25, 24, 32, 75, 28, 29, 38, 19, 19, 16, 28, 30, 29, 29, 19, 37, 34, 31, 35, 29, 19, 27, 42, 39, 11, 26, 21, 45, 61.

Select a systematic random sample of 10 units and find the average of it. 5

P. T. O.

[4]

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6. The yearly water consumption in a small city for ten years is given below. Determine the trend values by using a 3-yearly moving average. Also plot the values and interpret the results : 5

Year	Water consumption (in thousand gallon)
2007	26
2008	27
2009	28
2010	30
2011	29
2012	27
2013	30
2014	31
2015	32
2016	31

Section—B

7. Twenty individuals were randomly assigned to four different production processes. Production of units per hour for the four processes are given below :

Process I	33	30	28	29	30
Process II	33	35	30	38	34
Process III	28	38	30	34	30
Process IV	34	28	29	26	28

Determine if there is significant difference in the average production units per hour for the four types of production processes. [Given that $F_{(3,16)(5\%)} = 3.24$].

8. A population of size 10000 is divided into 4 strata. Their sizes and standard deviations are given below :

	Strata			
	I	II	III	IV
$N_i = \text{Sizes}$	5000	1000	2000	2000
s_i	25	10	15	20

P. T. O.

A stratified random sample of size 500 is to be drawn from this population. Determine the sizes of samples from these strata, in case :

- (i) Proportional allocation is used
(ii) Neyman's optimum allocation is used.

9. 1000 students at college level were graded according to their IQ levels and the economic condition of their parents, as follows :

Economic Condition	IQ Level		
	High	Low	Total
Rich	230	170	400
Poor	470	130	600
Total	700	300	1000

Test the hypothesis that IQ levels are independent of the economic conditions at 1% level of significance.

[Given that $\chi^2_{1(0.01)} = 6.63$]

10

10. A gynecologist recorded the blood pressure (BP) of her patients and collected the following data :

Age (in years)	BP (in mmHg)
23	65
24	60
25	62
26	70
28	70
29	73
31	75
35	83
40	90

Fit the regression line for the given data and estimate BP if age of the patient is 38 years.

10

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(BCA) (Revised)****Term-End Examination****June, 2022****BCS-040 : STATISTICAL TECHNIQUES***Time : 2 hours**Maximum Marks : 50***Note :**

- (i) Attempt both sections, i.e., Section A and Section B.
- (ii) Attempt any **four** questions from Section A.
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SECTION A

1. A random sample of 50 students of mathematics taken from a total of 200, showed a mean of 75 and a standard deviation of 10. Find 95% confidence limits for mean. (Given $Z = 1.96$) 5
2. Calculate the mean and standard deviation from the following data : 5

Marks	No. of Students
0 – 10	10
10 – 20	9
20 – 30	25
30 – 40	30
40 – 50	16
50 – 60	10

3. In order to find the correlation coefficient between two variables X and Y from 20 pairs of observations, the following calculations were made :

$$\Sigma X = 15, \Sigma Y = + 6, \Sigma XY = 50,$$

$$\Sigma X^2 = 61 \text{ and } \Sigma Y^2 = 90$$

Calculate the correlation coefficient and the slope of the regression line of Y on X.

5

4. Differentiate between Simple Random Sampling and Systematic Random Sampling.

5

5. List the advantages and disadvantages of using a sampling approach instead of a census approach for studying the characteristics of data.

5

SECTION B

6. Describe the following tests in detail : 10

- (a) Time series and its components
- (b) Chi-square test for independence of attributes

7. (a) Compute the three yearly moving average of the following data : 5

<i>Day</i>	<i>Sales</i>
1	45
2	46
3	48
4	47
5	58
6	58
7	51
8	52
9	53
10	45
11	51
12	61

- (b) An automobile tyre manufacturing company claims that the average life of a particular grade of tyre is more than 20,000 km when used under normal driving conditions. A random sample of 16 tyres was tested and mean and standard deviation of 22,000 km and 5,000 km respectively were calculated. Assuming the life of the tyres in km to be normally distributed, decide whether the manufacturer's claim is true ? Use 5% level of significance. (Given $t_{0.05, 15} = 1.75$) 5

8. The following contingency table presents the analysis of 300 persons according to hair colour and eye colour :

<i>Hair Colour</i>	<i>Eye Colour</i>		
	Blue	Grey	Brown
Fair	30	10	40
Brown	40	20	40
Black	50	30	40

Test the hypothesis that there is an association between hair colour and eye colour at 5% level of significance. (Given that $\chi^2_{0.05, 4} = 9.49$)

10

9. A computer engineer identifies A, B and C as three methods to do a certain job. To determine how long the operator takes to do the job when each of these methods is used, the engineer asks four operators to do the job using the method A, another four operators to do the job using the method B, and so on. Each operator's time (in minutes) is shown below :

A	B	C
19	18	21
17	16	20
22	15	19
20	14	19

Construct the relevant analysis of variance table and test the hypothesis that the average time of all operators are equal at 5% level of significance. (Given that $F_{0.05, (2, 9)} = 4.26$)

10

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BCS-040 : STATISTICAL TECHNIQUES

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Section—A

1. In a study on the per capita income for a particular year in a city, the following weekly observations were made : 5

Per Capita Income (₹) (1 k = 1000)	Number of Weeks
14 k–15 k	5
15 k–16 k	10
16 k–17 k	20
17 k–18 k	9
18 k–19 k	6
19 k–20 k	2

Draw a histogram and frequency polygon on the same scale.

2. A problem of statistics techniques is given to three students A, B and C whose chances of solving it are $\frac{1}{2}$, $\frac{1}{3}$ and $\frac{1}{4}$ respectively. 5

- (i) What is the probability that the problem will be solved ?
- (ii) What is the probability that only one of them will solve the problem correctly ?

3. Determine mean and median for the following data : 5

Marks	No. of Students
0—10	10
10—20	9
20—30	25
30—40	30
40—50	16
50—60	10

4. Box A contains 5 red and 4 blue balls, Box B contains 2 red and 5 blue balls. A ball is drawn at random from each box. Find the probability that one is red and the other is blue. 5
5. A statistics professor has given four tests. A student scored 75, 65, 80 and 95 respectively in the four tests. The professor decides to determine his grade by randomly selecting a sample of 2 test scores. Construct the sampling distribution for this process. 5

Section—B

6. Find and plot the regression line of y on x for the data given below : 10

Speed (km/hr) (x)	Stopping distance (in feet) (y)
30	160
40	240
50	330
60	435

7. In a partially destroyed laboratory, the legible record of analysis of correlation of data, is as follows : 10

Variance of $x = 9$, Regression equations :

- (i) $8x - 10y + 66 = 0$
 (ii) $4x - 18y - 214 = 0$

What were (a) the means of x and y , (b) the coefficient of correlation between x and y and (c) the standard deviation of y ?

8. (a) Compare simple random sampling with replacement and simple random sampling without replacement. 2
 (b) Define time series and discuss various components of time series. 4

(c) Write short notes on the following : 2+2

(i) t -test

(ii) Properties of good estimator

9. The table given below shows the relation between the performance of students in Statistics and Computer Sciences. Test the hypothesis that the performance in Statistics is independent of the performance in Computer Sciences using 5% level of significance.

(Given that $\chi^2_{0.05,4} = 9.49$).

10

		Computer Science		
		High Grade	Medium Grade	Low Grade
Statistics	High Grade	36	72	42
	Medium Grade	34	122	44
	Low Grade	50	56	44

BACHELOR OF COMPUTER

APPLICATIONS (BCA) (REVISED)

Term-End Examination

June, 2023

BCS-040 : STATISTICAL TECHNIQUES

Time : 2 Hours

Maximum Marks : 50

Note : (i) Attempt both Sections i.e. Section A and

Section B.

(ii) Attempt any **four** questions from

Section A.

(iii) Attempt any **three** questions from

Section B.

(iv) Use of non-scientific calculator is
allowed.

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Section—A

1. Write the median formula, and use it to complete the frequency distribution given below : 5

Class Interval (C.I.)	Frequency
10—20	12
20—30	30
30—40	?
40—50	65
50—60	?
60—70	25
70—80	18

Given, the median value of 200 observations is 46.

2. Fit a linear trend $y = a + bx$, to the data collected from a unit that manufactures shoes : 5

Month (x)	Demand (y)
1	46
2	56
3	54
4	43
5	57
6	56

3. If 2% of the items manufactured in a factory are defective, then find the probability that : 5

- (i) There are 3 defectives in a sample of 100
(ii) There is no defective in a sample of 50.

4. Calculate the correlation coefficient from the following data (determined from the 20 pairs of observations of two variables X and Y) : 5

$$\Sigma x = 15, \Sigma y = -6, \Sigma xy = 50, \Sigma x^2 = 61, \Sigma y^2 = 90$$

5. The mean weekly sales of mobile phones in different mobile stores was 146.3 mobile phones per store. After an advertisement campaign the mean weekly sales of 22 stores, increased to 153.7 (for a typical week), and showed a standard deviation of 17.2. Verify the success of advertisement campaign, at 5% level of significance. (Given : $t_{21(0.05)} = 2.08$). 5

6. The mobile phone numbers are combinations of ten digits, 0 to 9. An observer analysed the mobile numbers in the contact list of his/her mobile phone and prepared a frequency distribution table of the digits (0 to 9), as given ahead : 5

Digit	Frequency
0	99
1	100
2	82
3	65
4	50
5	77
6	88
7	57
8	82
9	30

From the observed data (given above), she/he wants to test whether the digits occur with same frequency or not. (Given : $\alpha = 0.05$), $\chi^2_{9(0.05)} = 16.918$).

Section—B

7. A survey study includes three villages V_1 , V_2 and V_3 having 50000, 30000 and 40000 as respective populations. A stratified random sample is to be taken with a total sample size of $n = 500$. Determine the sample size to be selected from each village individually using

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the method of (i) proportional allocation and (ii) optimal allocation.

From the previous survey, it is known that the standard deviations are $S_1 = 30$, $S_2 = 15$ and $S_3 = 20$. 10

8. The data of 300 persons, according to hair colour and eye colour is shown below : 10

	Eye Colour			
	Blue	Grey	Brown	
Hair Colour	White	30	10	40
	Brown	40	20	40
	Black	50	30	40

Test the hypothesis that there is association between hair colour and eye colour at 5% level of significance. (Given that : $\chi^2_{0.05,4} = 9.49$).

9. An engineer identifies four ways that a job can be done. To determine how long it takes for an operator to do a job, when each of these ways (or methods) are used, the engineer asks 4 operators to do a job using method A another 4 operators to do the same job using method B

P. T. O.

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and so on. Data, related to the performance of each operator (in seconds) is shown below : 10

A	B	C	D
19	18	21	22
17	16	20	23
22	15	19	21
20	14	19	20

Construct the relevant analysis of variance table and test the hypothesis that the average time of all operators are equal at 1% level of significance. (Given that : $F_{0.01, (3, 12)} = 5.95$).

10. Write short notes on any two of the following :

5×2=10

- (a) Control charts
- (b) Stratified sampling
- (c) Forecasting models

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**BACHELOR OF COMPUTER
APPLICATIONS (BCA) (REVISED)**

Term-End Examination

December, 2023

BCS-040 : STATISTICAL TECHNIQUES

Time : 2 Hours

Maximum Marks : 50

Note : (i) *Attempt both Sections i.e. Section A and Section B.*

(ii) *Attempt any **four** questions from Section A.*

(iii) *Attempt any **three** questions from Section B.*

(iv) *Use of non-scientific calculator is allowed.*

Section—A

1. (a) An electric bulb manufacturing company chooses a random sample of 10 bulbs, received from one of the suppliers. It

P. T. O.

determines life of each bulb. The result (in thousands of hours) are as follows :

4.5, 5.0, 4.2, 4.8, 4.2, 5.1, 4.0, 4.2, 4.2, 4.5

Compute and analyse a point estimate of the mean length of the life of the bulbs received from the supplier.

- (b) Compare parametric and non-parametric tests.

2. An insurance company has insured 1000 truck drivers, 3000 car drivers and 6000 scooter drivers. The probabilities that the truck, car and scooter drivers meet with an accident are 0.2, 0.04 and 0.25, respectively. One of the insured persons meets with an accident. What is the probability that the person is a car driver ?
3. A football manufacturing company wants to check the variation in the weight of balls. For this, 25 samples (each of size 4) are selected. The weight of each ball is measured (in grams), the sum of sample averages and sum of sample

ranges were found to be $\sum_{i=1}^{25} \bar{x}_i = 4010$ grams

and $\sum_{i=1}^{25} R_i = 72$ grams, respectively. 5

Compute the control limits for the $\bar{X}$ and R-chart. It is given that $A_2 = 0.729$, $D_3 = 0$ and $D_4 = 2.282$.

4. The frequency distribution of the accidental data of the factory for the last 50 weeks is shown below :

No. of Accidents	No. of Weeks
0—5	8
5—10	22
10—15	10
15—20	8
20—25	2

Draw the histogram and calculate the average number of accidents per week. 5

5. In order to test whether there is any significant difference between the proportion of safety

consciousness of men and women, while driving a car, a study was conducted. The study includes a sample of 300 men and 300 women. Out of 300 men, 130 said that they used seat belts, and out of 300 women, 90 said that they used seat belts. Based on the given data, test the claim that there is no significant difference between the proportion of safety consciousness of men and women, while driving a car at 5% level of significance. (Given that $Z_{0.025} = 1.96$). 5

6. A company manufactures two types of machines (A and B). The manager of the company tests a random sample of 50 machines of Type A and 60 machines of Type B and found the following information :

	Mean Life (in hours)	Standard Deviation (in hours)
Type A	1300	50
Type B	1200	60

Obtain 99% confidence interval for the difference of the average life of the two types of machines. (Given that $Z_{0.005} = 2.58$). 5

Section—B

7. To enforce the speed limit at four different locations in the city, the Police plans to install radar traps at each of the locations L_1 , L_2 , L_3 and L_4 . The radar traps at each of the locations L_1 , L_2 , L_3 and L_4 are operated 40%, 30%, 20% and 30% of the time. If a person who is speeding on his way to work has probabilities of 0.2, 0.1, 0.5 and 0.2 respectively, of passing through these locations, what is the probability that he will receive a speeding ticket ? Find also the probability that he will receive a speeding ticket at locations L_1 , L_2 , L_3 and L_4 . 10
8. Find and plot the regression line of y on x , for the data given below : 10

Speed (km/hr.)	30	40	50	60
Stopping distance (in feet)	160	240	330	435

9. A chemical firm wants to determine how four catalysts differ in yield ? The firm runs the experiment in three of its plant, namely A, B & C. In each plant, the yield is measured with each catalyst. The yield (in quintals) are as follows :

Catalyst Plant	1	2	3	4
A	2	1	2	4
B	3	2	1	3
C	1	3	3	1

Perform an ANOVA and comment whether the yield due to a particular catalyst is significant or not at 5% level of significance (Given $F_{3,6} = 4.76$). 10

10. In order to study the impact of air pollution on households, a random sample of 200 households was selected from each of the two communities. The respondent in each house was asked whether or not any one in the house was bothered by air pollution. The responses are

tabulated below (Given $\chi^2_{1(0.05)} = 3.841$,
 $\alpha = 0.05$) :

Community	Yes	No	Total
I	43	157	200
II	81	119	200
Total	124	276	400

Can the researcher conclude that the 2 communities are bothered differently by air pollution ?

10

**BACHELOR OF COMPUTER APPLICATIONS
(BCA) (Revised)****Term-End Examination****June, 2018****BCS-040 : STATISTICAL TECHNIQUES***Time : 2 hours**Maximum Marks : 50***Note :**

- (i) Attempt both Section A and Section B.
- (ii) Attempt any **four** questions from Section A.
- (iii) Attempt any **three** questions from Section B.

SECTION A

1. Draw a histogram and a frequency polygon for the following data. Also find its mean. Further comment on the shape of the distribution.

5

<i>Time (in minutes)</i>	<i>No. of Machines</i>
20 - 25	03
25 - 30	07
30 - 35	11
35 - 40	08
40 - 45	05
45 - 50	02

2. A sample of size 3 is to be selected from a population of 10 households. List all possible samples by (a) Linear systematic; (b) Circular systematic sampling. 5

3. A bag contains 10 white and 3 black balls. Balls are drawn one by one without replacement till all the black balls are drawn. Find the probability that this procedure comes to an end at the 6th draw. 5

4. From a list of 500 names and addresses, 100 names were selected without replacement and 25 wrong addresses were found. Identify the population and estimate the total number of addresses required to be corrected in the list. Also estimate the standard error. 5

5. In order to find the correlation coefficient between two variables X and Y from 20 pairs of observations, the following calculations were made :

$$\sum X = 15, \sum Y = -6, \sum XY = 50, \sum X^2 = 61 \\ \text{and } \sum Y^2 = 90.$$

Calculate the correlation coefficient and the slope of the regression line of Y on X. 5

6. A random sample of size 10 from a normal population gives the values 64, 72, 65, 70, 68, 71, 65, 62, 66, 67. If it is known that the standard error of the sample mean is $\sqrt{0.64}$, find 95% confidence limits for the population mean. Also find the population variance. 5

SECTION B

7. (a) The police plans to enforce speed limits by using radar traps at 4 different locations within the city limits. The radar traps at each of the locations L_1 , L_2 , L_3 and L_4 are operated 40%, 30%, 20% and 10% of the time respectively. If a person who is speeding on his way to work has probabilities 0.2, 0.1, 0.5 and 0.2 respectively, of passing through these locations, what is the probability that he will receive a speeding ticket? Also find the probability that he will receive a speeding ticket at location L_1 .

5

- (b) Explain the exponential smoothing method. Use the same to forecast the following data. Also estimate the error. [Take $W = 0.1$]

Table : Number of new colleges opened in the country.

5

Year	Colleges
2001	5
2002	3
2003	3
2004	4
2005	3
2006	6
2007	4

8. The result of a survey regarding radio listeners' preference for all different types of music are given in the following table with listeners classified by age group :

Types of Music Preferred	Age Group		
	19 – 25	26 – 35	Above 35
National Music	60	60	30
Foreign Music	190	264	96
Indifferent	50	76	74

Test at 5% level of significance whether the type of music is dependent on the age group. Given that

$$\chi^2_{4,5\%} = 9.488.$$

10

9. A company wants to test whether its 3 salesmen A, B and C have the same selling ability. Their records of sales (in ₹ '000) during various weeks of the last month are given below :

Salesman	A	B	C
I st week	16	22	25
II nd week	20	20	24
III rd week	18	15	16
IV th week	25	26	20

Prepare a one-way ANOVA table and test the hypothesis that all the salesmen perform equally over the week. Given $F_{5\%}(2, 9) = 4.26$.

10

10. Write short notes on any *four* of the following :

$$4 \times 2 \frac{1}{2} = 10$$

- (a) Cluster Sampling
 - (b) Stratified Sampling
 - (c) Control Charts
 - (d) Goodness of Fit
 - (e) Moving Average
-

**BACHELOR OF COMPUTER APPLICATIONS
(BCA) (Revised)**

Term-End Examination

December, 2018

02663

BCS-040 : STATISTICAL TECHNIQUES

Time : 2 hours

Maximum Marks : 50

Note :

- (i) Attempt both Section A and Section B.
- (ii) Attempt any **four** questions from Section A.
- (iii) Attempt any **three** questions from Section B.

SECTION A

1. The mean and standard deviation of 20 items is found to be 10 and 2, respectively. At the time of checking it was found that one item with value 8 was incorrect. Calculate the mean and standard deviation if the wrong item is omitted. 5
2. A population consists of the numbers 2, 5, 7, 8 and 10. Write all possible simple random samples of size 3 (without replacement). Verify that the sample mean is an unbiased estimator of the population mean. 5

3. A random sample of 10 males from a normal population showed a mean height of 66 inches and the sum of squares from this mean is equal to 90 sq inches. Is it reasonable to believe that the average height is greater than 64 inches ? Justify your answer at 5% level of significance (Use $t_{0.05}(9) = 2.26$). 5
4. Write short notes on any **two** of the following : $2 \times 2 \frac{1}{2} = 5$
- Simple Random Sampling
 - Systematic Sampling
 - Stratified Sampling
5. A random sample of size 64 has been drawn from a population with standard deviation 20. The mean of the sample is 80.
- Calculate 95% confidence limits for the population mean.
 - How does the width of the confidence interval change if the sample size is 256 instead ? 5
6. Box A contains 5 red and 4 blue balls, Box B contains 2 red and 5 blue balls. A ball is drawn at random from each box. Find the probability that one is red and the other is blue. 5

SECTION B

7. The following table gives data for a sample of married women, the level of education and marriage adjustment score :

Level of Education	Marriage Adjustment Score		
	Low	High	Very High
Middle School	25	5	10
High School	50	30	40
College	120	60	60

Can you conclude from the above, the higher the level of education, the greater is the adjustment in marriage ? Justify. Given that $\chi^2_{0.05}(4) = 9.488$. 10

8. Refill of cartons with apple juice is taking place in a plant. Data for 14 days was collected and 100 cartons were checked every day for proper filling. Data is as given below :

S. No.	X
1	8
2	2
3	4
4	1
5	3
6	3
7	2
8	4
9	9
10	7
11	5
12	8
13	5
14	9

Compute the UCL, LCL and CL using an appropriate control chart. Also draw the chart. 10

9. Plot the following data about demand for an item. Find the centred moving averages by taking $n = 3$. Use these to forecast the next two months' demand.

10

Month	Demand
1	46
2	56
3	54
4	43
5	57
6	56
7	67
8	62
9	50
10	56
11	47
12	56

10. The following table gives the yield of a hybrid variety of wheat in quintals per acre from 17 trial plots of land treated with four types of fertilizers :

Treatment with fertilizers

A	B	C	D
24	31	39	38
39	25	41	32
35	26	33	35
	21	40	34
		45	26

Use ANOVA to test whether the effect of fertilizers differs in terms of average yields.

Given that $F_{0.05}(3, 13) = 3.41$.

10

**BACHELOR OF COMPUTER APPLICATIONS
(BCA) (Revised)**

Term-End Examination

December, 2018

02663

BCS-040 : STATISTICAL TECHNIQUES

Time : 2 hours

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Note :

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5	3
6	3
7	2
8	4
9	9
10	7
11	5
12	8
13	5
14	9

Compute the UCL, LCL and CL using an appropriate control chart. Also draw the chart. 10

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	21	40	34
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Use ANOVA to test whether the effect of fertilizers differs in terms of average yields.

Given that $F_{0.05}(3, 13) = 3.41$.

10